```sql
use Renewable_Energy_India_Analysis

select * from capacity

select * from potential

Delete FROM capacity
WHERE State_UT = 'Total';

Delete FROM potential
WHERE State_UT = 'Total';

UPDATE capacity
SET State_UT = 'Odisha'
WHERE State_UT = 'Orissa';

UPDATE potential
SET State_UT = 'Pondicherry'
WHERE State_UT = 'Puducherry';

UPDATE capacity
SET State_UT = 'Jammu and Kashmir'
WHERE State_UT = 'Jammu & Kashmir';

UPDATE capacity
SET State_UT = 'Andaman and Nicobar Islands'
WHERE State_UT = 'Andaman & Nicobar Islands';

UPDATE capacity
SET State_UT = 'Dadra and Nagar Haveli and Daman and Diu'
WHERE State_UT = 'Dadra and Nagar Haveli and Daman & Diu';

UPDATE capacity
SET State_UT = 'Delhi'
WHERE State_UT = 'NCT of Delhi';

--1.
SELECT TOP 10
State_UT,
Total_RE_Capacity_MW
FROM capacity
ORDER BY Total_RE_Capacity_MW DESC;

--2.
SELECT
State_UT,
Solar_Potential_Total_MW,
Wind_Potential_MW,
(Solar_Potential_Total_MW - Wind_Potential_MW) AS Solar_Over_Wind_Diff_MW
```

```sql
FROM potential
WHERE Solar_Potential_Total_MW > Wind_Potential_MW
ORDER BY Solar_Over_Wind_Diff_MW DESC;

--3.
SELECT
State_UT,
Total_RE_Capacity_MW
FROM capacity
WHERE Total_RE_Capacity_MW > (SELECT AVG(Total_RE_Capacity_MW)
FROM capacity)
ORDER BY Total_RE_Capacity_MW DESC;

--4.
SELECT
c.State_UT,
c.Total_RE_Capacity_MW,
p.Total_RE_Potential_MW,
ROUND((c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 100, 2) AS
  Utilization,
CASE
WHEN (c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 100 >= 20.0
  THEN 'High'
WHEN (c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 100 >= 6.0
  THEN 'Moderate'
ELSE 'Low'
END AS RE_Utilization_Category
FROM capacity c
INNER JOIN potential p ON c.State_UT = p.State_UT
ORDER BY Utilization DESC;

--5
SELECT TOP 5
c.State_UT,
c.Total_RE_Capacity_MW,
p.Total_RE_Potential_MW,
ROUND((c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 10000, 2) AS
  Capacity_Per_10000_MW_Potential
FROM capacity AS c
INNER JOIN potential AS p ON c.State_UT = p.State_UT
ORDER BY Capacity_Per_10000_MW_Potential DESC;

--6
SELECT
c.State_UT,
ROUND(p.Total_RE_Potential_MW - c.Total_RE_Capacity_MW, 2) AS
  Untapped_RE_Potential_
  MW,                                          -- It shows how much
  room for expansion exists in a state before reaching its maximum
```

```
    theoretical clean energy limit.
92  ROUND((c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 100, 2) AS
    Utilization,                                                    -- A
    higher percentage indicates that the state has developed a large
    portion of its clean energy potential.
93  ROUND((c.Total_RE_Capacity_MW / (p.Total_RE_Potential_MW)) * 10000, 2) AS
    Capacity_Per_10k_Potential                                   -- higher is
    consider as has growth opportunity
94  FROM capacity AS c
95  FULL OUTER JOIN potential AS p ON c.State_UT = p.State_UT
96
97  --7
98  SELECT
99  c.State_UT,
100 p.Solar_Potential_Total_MW,
101 c.Solar_Capacity_MW
102 FROM capacity c
103 INNER JOIN potential p ON c.State_UT = p.State_UT
104 WHERE p.Solar_Potential_Total_MW > (SELECT AVG(Solar_Potential_Total_MW)
105 FROM potential) AND c.Solar_Capacity_MW < (SELECT AVG(Solar_Capacity_MW)
106 FROM capacity)
107 ORDER BY p.Solar_Potential_Total_MW DESC;
108
109
```